Data Address Map, Peripheral Bit Fields and Timer Control

WM-8 five-stage pipelined washing-machine microcontroller – ELL7282/ELL782 Introduction to Computer Architecture – Assignment 2

1. Address map (one page summary)

Address	Name	Access	Reset	Bit fields / behaviour
0x00–0x7F	RAM	R/W	undefined†	128×8 data RAM (Logisim RAM, separate data-in/out). Write when WE∧¬A7; asynchronous read.
0x80–0xEF	reserved	–	–	Reads return 0x00; writes are ignored (no RAM aliasing: A7=1 blocks RAM WE).
0xF0	IN_STATUS	R	live	b0 START | b1 DOOR_CLOSED | b3:2 MODE[1:0] | b4 TRIP (safety latch) | b5 ACTUAL_MOTOR_EN | b7:6 = 0. Writes ignored.
0xF1	WATER_LEVEL	R	live	b7:0 unsigned water level. Writes ignored.
0xF2	TEMP	R	live	b7:0 temperature read-back (sensor only). Writes ignored.
0xF3	DRUM_RPM	R	live	b7:0 drum-speed read-back (sensor only). Writes ignored.
0xF4	ACT (actuator latch)	R/W	0x00	b0 VALVE_IN | b1 HEATER | b2 MOTOR_EN (request) | b3 PUMP_OUT | b4 MOTOR_DIR | b5 DOOR_LOCK | b6 BUZZER | b7 reserved (write ignored, reads 0). Read-back returns the latch.
0xF5	DISPLAY	R/W	0x00	b7:0 status code / LED pattern, read-back returns the latch.
0xF6, 0xF7	unimplemented	–	–	read 0x00, writes ignored.
0xF8	T_COUNT	R	0x00	b7:0 current timer count. Writes ignored (cleared only via T_CTRL.CLR).
0xF9	T_LIMIT	R/W	0x00	b7:0 compare value; 0x00 disables comparison. Writing it never sets MATCH by itself.
0xFA	T_CTRL	R/W	0x00	b0 EN (R/W enable state) | b1 CLR (write-1 command: clear COUNT and MATCH; self-clearing, reads 0) | b7:2 ignored, read 0. One store sets EN:=b0 and, if b1=1, clears.
0xFB	T_STATUS	R	0x00	b0 MATCH (sticky) | b7:1 = 0. Writes ignored.
0xFC–0xFF	unimplemented	–	–	read 0x00, writes ignored.

†RAM and R0–R7 are not cleared by RESET; software initialises them before use (Logisim happens to power them up as 0).

2. Bus timing and write rules

- The MEM stage drives D_ADDR = EX/MEM.ALU (the forwarded Rs value), D_WDATA = EX/MEM.SDATA (the forwarded Rd value) and the single write strobe D_WE = EX/MEM.V∧EX/MEM.MW. Every peripheral register and the RAM write on the global rising clock edge that ends the MEM cycle, and only if D_WE=1. Each instruction spends exactly one cycle in MEM, because MEM is never stalled, so a store commits *exactly once*. Squashed instructions and bubbles have V=0, so they can never change RAM, actuators, display or timer.
- Decode: IS_RAM=¬A7; IS_PER=A7∧A6∧A5∧A4; a gate-level 4:16 decoder on A3:0 gives SEL_Fx=IS_PER∧dec[x]. Write enables: RAM_WE=D_WE∧IS_RAM, WE_ACT=D_WE∧SEL_F4, WE_DISP=D_WE∧SEL_F5, WE_LIMIT=D_WE∧SEL_F9, WE_CTRL=D_WE∧SEL_FA. No other address has a write enable, so read-only, reserved and unimplemented addresses ignore writes by construction.
- Read data = OR of 11 AND-gated sources (RAM, F0–F5, F8–FB). The selects are mutually exclusive, so an address with no select returns 0x00. Reads have no side effects. A load samples the inputs combinationally and the value is captured into MEM/WB at the edge ending its MEM cycle, so tests must hold inputs stable through that edge.
- MODE is snapshot by software: the program copies IN_STATUS[3:2] from the same LD that saw START∧DOOR_CLOSED into RAM 0x04 and register parameters, and never reads MODE again during a cycle.

3. Timer (0xF8–0xFB)

Registers COUNT, LIMIT, EN, MATCH are clocked by the *global* clock every rising edge and do not depend on CPU stall, flush or halt signals. INC=COUNT+1 comes from a gate-level half-adder chain. CLRCMD=WE_CTRL∧WDATA[1]. At each rising edge:

```
EN     ← WDATA[0] if WE_CTRL, else unchanged
COUNT  ← 0 if CLRCMD; else INC if EN; else unchanged        (clear has priority)
MATCH  ← 0 if CLRCMD; else MATCH ∨ (EN ∧ INC=LIMIT ∧ LIMIT≠0)
LIMIT  ← WDATA if WE_LIMIT, else unchanged
```

MATCH is therefore set only by an enabled increment whose *new* count equals a non-zero LIMIT, using the values held before the edge. It stays set until a clear. Writing LIMIT alone never sets it. COUNT wraps 0xFF→0x00. RESET (asynchronous register clear) gives COUNT=LIMIT=0, EN=MATCH=0. **Software recipes:** start/re-arm an interval of N ticks: ST N,[F9] then ST 0x03,[FA] (clear + enable). MATCH becomes 1 on the N-th edge after the CTRL store. Pause: ST 0x01/0x00,[FA]. Stop and clear: ST 0x02,[FA]. Poll LD Rx,[FB].

4. Safety-gated outputs

ActualMotorEnable = ACT[2] ∧ DOOR_CLOSED ∧ ¬TRIP, computed by gates outside the CPU. It appears on the MOTOR_EN_ACTUAL pin and in IN_STATUS[5]. The commanded request stays visible on MOTOR_EN. TRIP (in IN_STATUS[4]) is set at a clock edge where the door is sampled open while ACT[2]=1. It clears only at an edge with ACT[2]=0, DOOR_CLOSED=1 and a new START 0→1 transition (START∧¬START_prev), or on RESET. All other actuator outputs equal their latch bits.